

Python Programming for Business Analytics

Final Project

Author: KAO, Ting-Yi

Import Essential Packages

```
In [72]: # 🚀 Run this cell first to set up your environment!
# Data Handling and Analysis Libraries
import pandas as pd
import numpy as np
# Visualization Libraries
import matplotlib.pyplot as plt
import seaborn as sns
import plotly as py
# Mathematical and Statistical Libraries
import scipy as sp

# For nicer plots
plt.style.use('seaborn-v0_8-whitegrid')
plt.rcParams['figure.figsize'] = (10, 6)

print("✅ Libraries loaded successfully!")
print(f"📊 Pandas version: {pd.__version__}")
print(f"🔢 NumPy version: {np.__version__}")
print(f"🎨 Seaborn version: {sns.__version__}")
print(f"🌐 Plotly version: {py.__version__}")
print(f"📦 SciPy version: {sp.__version__}")
print("\n🎓 Ready to learn! Let's go!")
```

✅ Libraries loaded successfully!

📊 Pandas version: 2.3.2

🔢 NumPy version: 2.2.5

🎨 Seaborn version: 0.13.2

🌐 Plotly version: 6.5.2

📦 SciPy version: 1.15.3

🎓 Ready to learn! Let's go!

Section 1 - Introduction

1.1 Motivation

In the era of rapid globalization and technological advancement, the disparity in human development across different regions remains a critical subject of study. As a student transitioning dig into Data Analysis, I am motivated to explore how historical economic

shifts correlate with tangible improvements in human well-being. The **Gapminder dataset** provides a chance to examine these trends, revealing the data-driven reality of our world's evolution over the last half-century.

1.2 Research Purpose & Questions

The primary objective of this project is to conduct a multi-dimensional analysis of the relationship between economic output (**GDP per Capita**) and public health outcomes (**Life Expectancy**). By leveraging Python's analytical ecosystem, I aim to move from basic observation to quantified insights.

This analysis will specifically address the following research questions:

1. **The Wealth-Health Correlation:** How strong is the statistical relationship between a nation's wealth and the longevity of its citizens, and does this relationship hold true across all continents?
2. **Temporal Evolution:** How has the global distribution of health and wealth shifted between 1952 and 2007?
3. **Regional Divergence:** Which geographic regions have shown the most significant "catch-up" effect in terms of life expectancy, regardless of their economic starting point?

1.3 Analytical Roadmap

To provide a comprehensive answer to these questions, the project will follow a structured data science workflow:

- **Data Preparation:**
Cleaning and transforming the Gapminder data, including log transformations for skewed economic metrics.
- **Exploratory Data Analysis (EDA):**
Using statistical summaries and grouping to identify global trends.
- **Machine Learning & Statistics:**
Applying Linear Regression to model the relationship between wealth and health, alongside Principal Component Analysis (PCA) to explore the variance and underlying structure of the global development metrics.
- **Advanced Visualization:**
Implementing interactive charts to visualize the 55-year trajectory of global development.
- **Conclusion:**
Concluding findings to provide actionable insights into the current state of global

inequality.

Section 2 - Data Exploration

2.1 Basic Structural Inspection

Before deep-diving into analysis, I will verify the dataset's integrity and dimensions.

```
In [73]: # Importing the dataset
df = pd.read_csv('https://raw.githubusercontent.com/plotly/datasets/master/gdpPerCapitaInequality.csv')
```

```
Out[73]:
```

	country	continent	year	lifeExp	pop	gdpPercap	iso_alpha	iso_nui
0	Afghanistan	Asia	1952	28.801	8425333	779.445314	AFG	
1	Afghanistan	Asia	1957	30.332	9240934	820.853030	AFG	
2	Afghanistan	Asia	1962	31.997	10267083	853.100710	AFG	
3	Afghanistan	Asia	1967	34.020	11537966	836.197138	AFG	
4	Afghanistan	Asia	1972	36.088	13079460	739.981106	AFG	
...	...	...	...	...	...	...	...	...
1699	Zimbabwe	Africa	1987	62.351	9216418	706.157306	ZWE	71
1700	Zimbabwe	Africa	1992	60.377	10704340	693.420786	ZWE	71
1701	Zimbabwe	Africa	1997	46.809	11404948	792.449960	ZWE	71
1702	Zimbabwe	Africa	2002	39.989	11926563	672.038623	ZWE	71
1703	Zimbabwe	Africa	2007	43.487	12311143	469.709298	ZWE	71

1704 rows x 8 columns

```
In [74]: # Check dimensions
print(f"Dataset Shape: {df.shape}")

# Inspect first few rows
display(df.head())

# Check data types and missing values
df.info()
```

Dataset Shape: (1704, 8)

	country	continent	year	lifeExp	pop	gdpPercap	iso_alpha	iso_num
0	Afghanistan	Asia	1952	28.801	8425333	779.445314	AFG	4
1	Afghanistan	Asia	1957	30.332	9240934	820.853030	AFG	4
2	Afghanistan	Asia	1962	31.997	10267083	853.100710	AFG	4
3	Afghanistan	Asia	1967	34.020	11537966	836.197138	AFG	4
4	Afghanistan	Asia	1972	36.088	13079460	739.981106	AFG	4

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 1704 entries, 0 to 1703
```

```
Data columns (total 8 columns):
```

#	Column	Non-Null Count	Dtype
0	country	1704 non-null	object
1	continent	1704 non-null	object
2	year	1704 non-null	int64
3	lifeExp	1704 non-null	float64
4	pop	1704 non-null	int64
5	gdpPercap	1704 non-null	float64
6	iso_alpha	1704 non-null	object
7	iso_num	1704 non-null	int64

```
dtypes: float64(2), int64(3), object(3)
```

```
memory usage: 106.6+ KB
```

The dataset contains 1704 records and 8 columns, and no missing values were observed. However, further research is required.

```
In [75]: # print country names
print("Countries in the dataset:")
print(df['country'].unique())

# print years covered
print("Years covered in the dataset:")
print(df['year'].unique())
```

Countries in the dataset:

['Afghanistan' 'Albania' 'Algeria' 'Angola' 'Argentina' 'Australia'
'Austria' 'Bahrain' 'Bangladesh' 'Belgium' 'Benin' 'Bolivia'
'Bosnia and Herzegovina' 'Botswana' 'Brazil' 'Bulgaria' 'Burkina Faso'
'Burundi' 'Cambodia' 'Cameroon' 'Canada' 'Central African Republic'
'Chad' 'Chile' 'China' 'Colombia' 'Comoros' 'Congo, Dem. Rep.'
'Congo, Rep.' 'Costa Rica' 'Cote d'Ivoire' 'Croatia' 'Cuba'
'Czech Republic' 'Denmark' 'Djibouti' 'Dominican Republic' 'Ecuador'
'Egypt' 'El Salvador' 'Equatorial Guinea' 'Eritrea' 'Ethiopia' 'Finland'
'France' 'Gabon' 'Gambia' 'Germany' 'Ghana' 'Greece' 'Guatemala' 'Guinea'
'Guinea-Bissau' 'Haiti' 'Honduras' 'Hong Kong, China' 'Hungary' 'Iceland'
'India' 'Indonesia' 'Iran' 'Iraq' 'Ireland' 'Israel' 'Italy' 'Jamaica'
'Japan' 'Jordan' 'Kenya' 'Korea, Dem. Rep.' 'Korea, Rep.' 'Kuwait'
'Lebanon' 'Lesotho' 'Liberia' 'Libya' 'Madagascar' 'Malawi' 'Malaysia'
'Mali' 'Mauritania' 'Mauritius' 'Mexico' 'Mongolia' 'Montenegro'
'Morocco' 'Mozambique' 'Myanmar' 'Namibia' 'Nepal' 'Netherlands'
'New Zealand' 'Nicaragua' 'Niger' 'Nigeria' 'Norway' 'Oman' 'Pakistan'
'Panama' 'Paraguay' 'Peru' 'Philippines' 'Poland' 'Portugal'
'Puerto Rico' 'Reunion' 'Romania' 'Rwanda' 'Sao Tome and Principe'
'Saudi Arabia' 'Senegal' 'Serbia' 'Sierra Leone' 'Singapore'
'Slovak Republic' 'Slovenia' 'Somalia' 'South Africa' 'Spain' 'Sri Lanka'
'Sudan' 'Swaziland' 'Sweden' 'Switzerland' 'Syria' 'Taiwan' 'Tanzania'
'Thailand' 'Togo' 'Trinidad and Tobago' 'Tunisia' 'Turkey' 'Uganda'
'United Kingdom' 'United States' 'Uruguay' 'Venezuela' 'Vietnam'
'West Bank and Gaza' 'Yemen, Rep.' 'Zambia' 'Zimbabwe']

Years covered in the dataset:

[1952 1957 1962 1967 1972 1977 1982 1987 1992 1997 2002 2007]

Data Dictionary

To interpret the analysis correctly, it is vital to define the variables within the Gapminder dataset. The dataset contains both categorical and numerical features that track global development metrics:

Column Name	Data Type	Description
country	Nominal (Object)	The name of the specific country or territory (e.g., Afghanistan, Zimbabwe).
continent	Nominal (Object)	The geographic region the country belongs to (Africa, Americas, Asia, Europe, or Oceania).
year	Discrete (Int)	The year the data was recorded, ranging from 1952 to 2007 in 5-year increments.
lifeExp	Continuous (Float)	Life Expectancy: The average number of years a newborn is expected to live if mortality patterns remain constant.
pop	Discrete (Int)	Population: The total number of people living in the country during that specific year.
gdpPercap	Continuous (Float)	GDP per Capita: Gross Domestic Product divided by the total population, adjusted for inflation (measured in international dollars).

Column Name	Data Type	Description
iso_alpha	Nominal (Object)	The 3-letter international standard country code (e.g., AFG for Afghanistan).
iso_num	Nominal (Int)	The numerical international standard country code.

Key Analytical Note: In this project, **GDP per Capita** serves as the primary proxy for economic wealth, while **Life Expectancy** serves as the primary metric for public health outcomes.

2.2 Custom Summary & Distribution Analysis

To go beyond basic descriptive statistics, I will examine the distribution of our key variables across different continents. In addition, analyzing skewness across the entire dataset can be misleading due to global shifts over time. Here, I calculate the skewness for each year to observe how the distribution of health and wealth has evolved.

```
In [76]: # Basic numerical summary
display(df.describe())

# Segmented Summary: Life Expectancy by Continent (the most recent year 2007)
latest_year = df[df['year'] == 2007]
continent_summary = latest_year.groupby('continent')['lifeExp'].agg(['mean',
display(continent_summary)
```

	year	lifeExp	pop	gdpPercap	iso_num
count	1704.00000	1704.000000	1.704000e+03	1704.000000	1704.000000
mean	1979.50000	59.474439	2.960121e+07	7215.327081	425.880282
std	17.26533	12.917107	1.061579e+08	9857.454543	248.305709
min	1952.00000	23.599000	6.001100e+04	241.165876	4.000000
25%	1965.75000	48.198000	2.793664e+06	1202.060309	208.000000
50%	1979.50000	60.712500	7.023596e+06	3531.846988	410.000000
75%	1993.25000	70.845500	1.958522e+07	9325.462346	638.000000
max	2007.00000	82.603000	1.318683e+09	113523.132900	894.000000

	mean	median	std	min	max
continent					
Africa	54.806038	52.9265	9.630781	39.613	76.442
Americas	73.608120	72.8990	4.440948	60.916	80.653
Asia	70.728485	72.3960	7.963724	43.828	82.603
Europe	77.648600	78.6085	2.979813	71.777	81.757
Oceania	80.719500	80.7195	0.729027	80.204	81.235

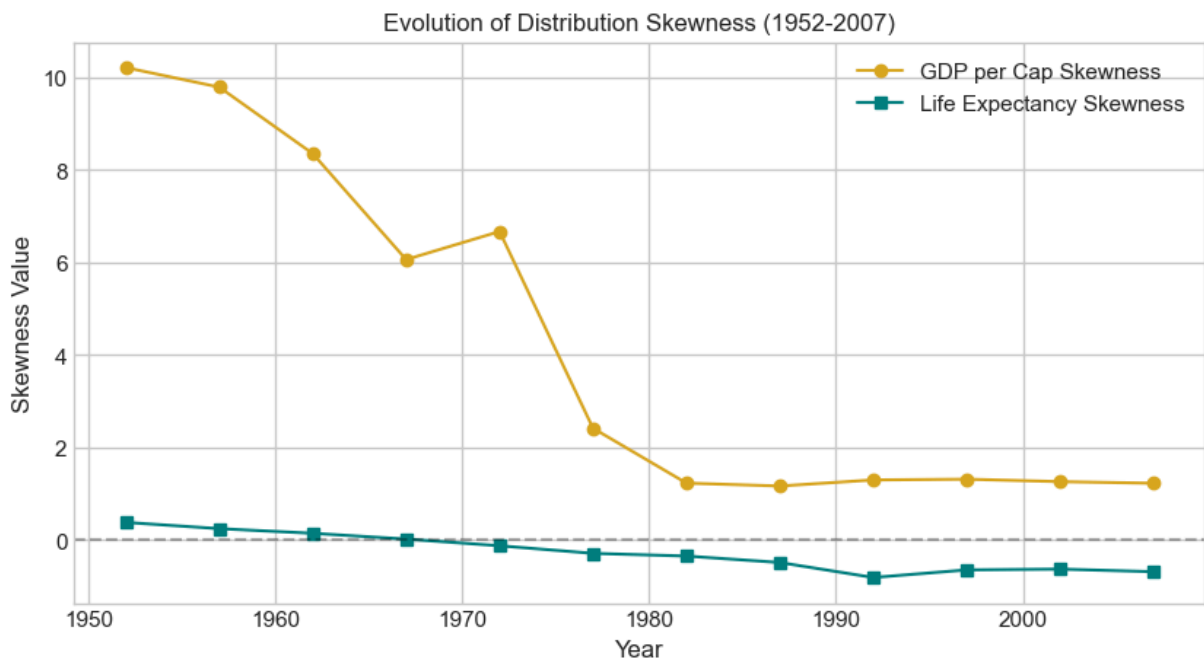
```
In [77]: # Calculate skewness for each year
skew_by_year = df.groupby('year')[['gdpPercap', 'lifeExp']].skew()

# Display the skewness trends
print("Skewness Trends over Time:")
display(skew_by_year)

# Plotting the trend to identify patterns
plt.figure(figsize=(10, 5))
plt.plot(skew_by_year.index, skew_by_year['gdpPercap'], marker='o', label='GDP Per Capita')
plt.plot(skew_by_year.index, skew_by_year['lifeExp'], marker='s', label='Life Expectancy')
plt.title('Evolution of Distribution Skewness (1952-2007)')
plt.xlabel('Year')
plt.ylabel('Skewness Value')
plt.legend()
plt.axhline(0, color='black', linestyle='--', alpha=0.3)
plt.show()
```

Skewness Trends over Time:

	gdpPercap	lifeExp
year		
1952	10.212219	0.377411
1957	9.789824	0.242491
1962	8.352427	0.142067
1967	6.061076	0.016697
1972	6.668169	-0.127588
1977	2.413530	-0.291895
1982	1.226803	-0.350068
1987	1.164286	-0.489871
1992	1.295808	-0.814135
1997	1.311835	-0.649674
2002	1.258700	-0.632298
2007	1.224198	-0.688777



2.3 Pattern Identification

By visualizing skewness over time rather than as a static aggregate, the following critical shifts in global development were found:

1. The GDP Wealth Compression:

- **Observation:** GDP per capita skewness was extremely high (~10) in 1952, indicating a world where wealth was concentrated in a tiny fraction of nations.

- **Trend:** There is a dramatic "compression" of wealth skewness between 1970 and 1980, followed by stabilization around a value between 1 and 2.
- **Insight:** While wealth is still positively skewed (right-tailed), the extreme "outlier wealth" of the post-WW2 era has transitioned into a more broad-based (though still unequal) global economy.

2. The Life Expectancy "Left-Behind" Shift:

- **Observation:** Life expectancy began with a slightly positive skew but crossed the zero-line around 1970, becoming increasingly **negatively skewed**.
- **Insight:** A negative skew indicates that the "tail" is now on the left. While the majority of the world is reaching higher age brackets, a small group of nations is being "left behind" with significantly lower life expectancy, likely due to regional crises or health epidemics.

3. Convergence of Skewness:

- Both metrics appear to be stabilizing in the most recent decades (1990–2007), suggesting that the "rules" of global health and wealth distribution have entered a new, more consistent phase.

Section 3 - Data Cleaning

In this section, I perform a cleanup and transformation of the data. While the initial inspection showed no null values, "cleaning" also involves addressing redundancy, ensuring data consistency, and preparing variables for statistical modeling.

3.1 Procedure Checklist

1. **Redundancy & Dimensionality Reduction:** Identify and remove redundant columns to streamline the dataset.
2. **Consistency Check:** Verify that categorical labels (country and continent names) are consistent and free of trailing spaces or duplicates.
3. **Handling Missing Values:** Although no nulls were found, I will implement a backup Check function to confirm integrity after any transformations.
4. **Feature Engineering (Transformation):** Apply a Logarithmic Transformation to the GDP per capita data to mitigate the extreme positive skew identified in the EDA.
5. **Outlier Treatment:** Identify and document extreme outliers in Life Expectancy that contribute to the "negative tail" found in the skewness analysis.

```
In [78]: # 1. Dropping redundant columns
df_clean = df.drop(columns=['iso_alpha', 'iso_num'])

# 2. Regex Cleaning & Consistency Check
# Ensure no leading/trailing spaces in string columns
df_clean['country'] = df_clean['country'].str.strip()
```

```

df_clean['continent'] = df_clean['continent'].str.strip()

# 3. Sophisticated Transformation: Log-Scaling
# Based on the skewness analysis, applying a log10 transformation to GDP
df_clean['log_gdpPercap'] = np.log10(df_clean['gdpPercap'])

# 4. Outlier Detection Function (Quantified Insight)
def detect_health_outliers(data, threshold=2):
    """
    Identifies countries where Life Expectancy is significantly lower than
    the mean for that specific year.
    """
    mean = data['lifeExp'].mean()
    std = data['lifeExp'].std()
    return data[data['lifeExp'] < (mean - threshold * std)]

# Flagging the 'Negative Tail' outliers in the most recent year (2007)
health_outliers = detect_health_outliers(df_clean[df_clean['year'] == 2007])
print(f"Significant Life Expectancy Outliers (2007):\n")
display(health_outliers[['country', 'lifeExp', 'log_gdpPercap']])

```

Significant Life Expectancy Outliers (2007):

	country	lifeExp	log_gdpPercap
47	Angola	42.731	3.680991
887	Lesotho	42.592	3.195715
1043	Mozambique	42.082	2.915761
1355	Sierra Leone	42.568	2.935780
1463	Swaziland	39.613	3.654512
1691	Zambia	42.384	3.104218

```

In [79]: # Summary the cleaned dataset
print("Cleaned Dataset Summary:")
df_clean.info()
display(df_clean.describe())
df_clean.groupby('year')[['log_gdpPercap', 'lifeExp']].skew()

```

```

Cleaned Dataset Summary:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1704 entries, 0 to 1703
Data columns (total 7 columns):
#   Column                Non-Null Count  Dtype
---  -
0   country                1704 non-null   object
1   continent              1704 non-null   object
2   year                   1704 non-null   int64
3   lifeExp                1704 non-null   float64
4   pop                    1704 non-null   int64
5   gdpPercap              1704 non-null   float64
6   log_gdpPercap          1704 non-null   float64
dtypes: float64(3), int64(2), object(2)
memory usage: 93.3+ KB

```

	year	lifeExp	pop	gdpPercap	log_gdpPercap
count	1704.00000	1704.000000	1.704000e+03	1704.000000	1704.000000
mean	1979.50000	59.474439	2.960121e+07	7215.327081	3.543318
std	17.26533	12.917107	1.061579e+08	9857.454543	0.539031
min	1952.00000	23.599000	6.001100e+04	241.165876	2.382316
25%	1965.75000	48.198000	2.793664e+06	1202.060309	3.079926
50%	1979.50000	60.712500	7.023596e+06	3531.846988	3.548002
75%	1993.25000	70.845500	1.958522e+07	9325.462346	3.969670
max	2007.00000	82.603000	1.318683e+09	113523.132900	5.055084

Out[79]:

	log_gdpPercap	lifeExp
year		
1952	0.362748	0.377411
1957	0.317023	0.242491
1962	0.213507	0.142067
1967	0.135591	0.016697
1972	0.081263	-0.127588
1977	-0.030442	-0.291895
1982	-0.100478	-0.350068
1987	-0.089121	-0.489871
1992	-0.032181	-0.814135
1997	-0.054895	-0.649674
2002	-0.070714	-0.632298
2007	-0.154052	-0.688777

3.2 Feature Selection & Redundancy Problem

The dataset contains 8 columns, but not all are necessary for a high-quality analysis. I have made the following decisions regarding column retention:

Column	Decision	Reason
country / continent	Keep	Essential categorical identifiers for regional grouping.
year / pop	Keep	Required for longitudinal analysis and weighting.
lifeExp / gdpPercap	Keep	Our primary dependent and independent variables.
iso_alpha / iso_num	Drop	Redundant. These are simply codes for the country column. Retaining them adds no analytical value and increases dimensionality.

Documenting Alternatives:

I considered keeping `iso_alpha` for potential use in geographic mapping (Choropleth maps). However, since Plotly can map data directly using country names, I have opted to drop them to maintain a clean dataset.

Section 4 - Data Analysis

In this section, I apply statistical methods and linear modeling to answer the core research questions.

4.1 RQ1: The Wealth-Health Correlation

First, I examine the strength of the linear relationship between our transformed economic metric (`log_gdpPercap`) and health outcomes (`lifeExp`).

```
In [80]: # 1. Correlation Analysis (Global)
correlation = df_clean[['lifeExp', 'log_gdpPercap']].corr()
print("Global Correlation Matrix:")
display(correlation)

# 2. Linear Regression Model (Quantifying the Relationship)
# We calculate the slope and intercept to write the equation: LifeExp = slope
slope, intercept, r_value, p_value, std_err = sp.stats.linregress(df_clean['
print(f"Regression Results:")
print(f"Slope (m): {slope:.2f}")
print(f"Intercept (b): {intercept:.2f}")
print(f"R-squared: {r_value**2:.4f}")
print(f"P-value: {p_value:.4e}")
```

```
print(f"\nModel Interpretation: For every 10x increase in GDP (1 unit log increase), Life Expectancy increases by approximately 19.4 years.")
```

Global Correlation Matrix:

	lifeExp	log_gdpPercap
lifeExp	1.000000	0.807618
log_gdpPercap	0.807618	1.000000

Regression Results:

Slope (m): 19.35

Intercept (b): -9.10

R-squared: 0.6522

P-value: 0.0000e+00

Model Interpretation: For every 10x increase in GDP (1 unit log increase), Life Expectancy increases by approximately 19.4 years.

4.2 RQ2: Temporal Evolution (Weighted Analysis)

Standard averages can be misleading because they treat small and large countries equally. Here, I perform a **Population-Weighted Analysis** to understand the "human" experience of progress versus the "national" experience.

```
In [81]: # 3. Custom Function: Weighted Mean
def weighted_mean(x):
    # Returns sum(value * weight) / sum(weight)
    return np.average(x['lifeExp'], weights=x['pop'])

# Groupby Year and apply custom aggregation
global_trend = df_clean.groupby('year').apply(weighted_mean).reset_index(name='weighted_mean')

# Compare with simple mean to see the difference
global_trend['simple_mean'] = df_clean.groupby('year')['lifeExp'].mean().values

print("Global Life Expectancy: Simple vs. Weighted Average")
display(global_trend.head())
```

Global Life Expectancy: Simple vs. Weighted Average

/var/folders/fl/f_hjv3lj25z58v5bn3btbssh0000gn/T/ipykernel_11928/3568382735.py:7: FutureWarning:

DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the operation. Either pass `include\_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.

	year	weighted_lifeExp	simple_mean
0	1952	48.944237	49.057620
1	1957	52.121885	51.507401
2	1962	52.324382	53.609249
3	1967	56.984307	55.678290
4	1972	59.514783	57.647386

4.3 RQ3: Regional Divergence (The "Catch-Up" Effect)

Which regions have improved the most? I use a pivot table to compare the starting line (1952) with the finish line (2007).

```
In [82]: # 4. Pivot Table Analysis
regional_progress = df_clean.pivot_table(
    index='continent',
    columns='year',
    values='lifeExp',
    aggfunc='median'
)[[1952, 2007]]

# 5. Feature Engineering: Calculate Growth
regional_progress['growth_years'] = regional_progress[2007] - regional_progress[1952]
regional_progress['growth_pct'] = (regional_progress['growth_years'] / regional_progress[1952]) * 100

print("Regional Development Velocity (1952-2007):")
display(regional_progress.sort_values('growth_years', ascending=False))
```

Regional Development Velocity (1952-2007):

	year	1952	2007	growth_years	growth_pct
continent					
	Asia	44.869	72.3960	27.5270	61.349707
	Americas	54.745	72.8990	18.1540	33.161019
	Africa	38.833	52.9265	14.0935	36.292586
	Europe	65.900	78.6085	12.7085	19.284522
	Oceania	69.255	80.7195	11.4645	16.554039

4.4 Statistical Hypothesis Testing

Is the gap between "Rich" and "Poor" nations statistically significant, or just due to chance?

- **Null Hypothesis (H_0):** There is no significant difference in Life Expectancy between High-Income and Low-Income countries.
- **Alternative Hypothesis (H_1):** High-Income countries have significantly higher Life Expectancy.

```
In [83]: # Define "Rich" and "Poor" based on the median GDP of 2007
data_2007 = df_clean[df_clean['year'] == 2007]
median_gdp = data_2007['gdpPercap'].median()

rich_countries = data_2007[data_2007['gdpPercap'] > median_gdp]['lifeExp']
poor_countries = data_2007[data_2007['gdpPercap'] <= median_gdp]['lifeExp']

# 6. Independent T-Test
t_stat, p_val = sp.stats.ttest_ind(rich_countries, poor_countries, equal_var=False)

print(f"T-Statistic: {t_stat:.4f}")
print(f"P-Value: {p_val:.4e}")

if p_val < 0.05:
    print("Result: Reject Null Hypothesis. The wealth gap statistically impacts health outcomes.")
else:
    print("Result: Fail to reject Null Hypothesis.")
```

T-Statistic: 11.3489

P-Value: 1.1725e-20

Result: Reject Null Hypothesis. The wealth gap statistically impacts health outcomes.

Section 5 - Visualization

Visualizing the analytical results allows us to identify non-linear patterns and regional outliers that summary statistics might miss. Here, I utilize **Plotly** for interactive data exploration and **Seaborn** for more advanced comparisons.

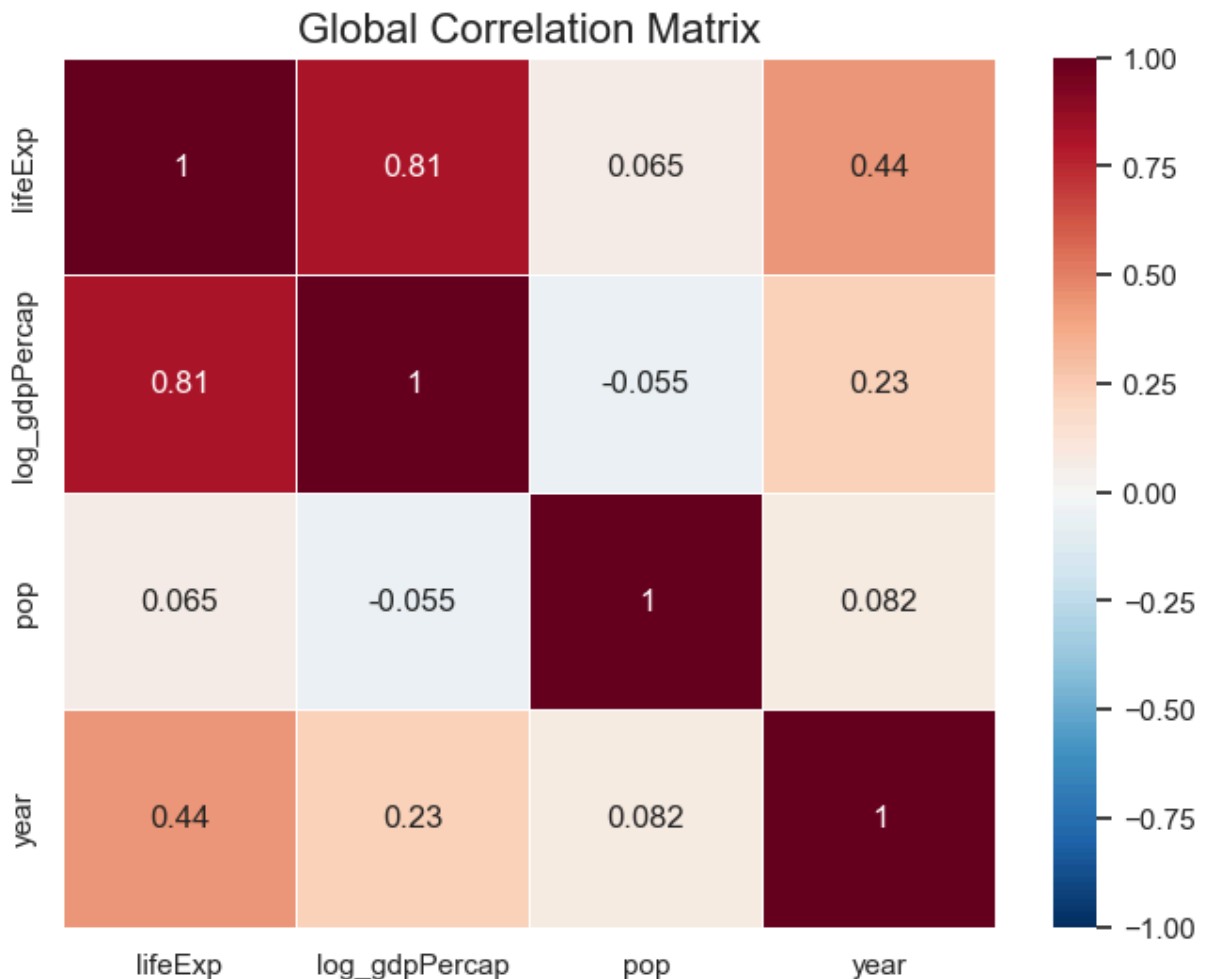
```
In [84]: # 5.1 Visualization Setup
import plotly.express as px
import plotly.graph_objects as go
import matplotlib.pyplot as plt
import seaborn as sns

# Set a professional color scheme
sns.set_theme(style="whitegrid", palette="muted")
plt.rcParams['figure.figsize'] = (12, 7)
plt.rcParams['font.size'] = 12
```

5.2 RQ1: The Wealth-Health Connection

A correlation of **0.81** was observed in Section 4. The following charts visualize this linear relationship and how it varies by region.

```
In [85]: # Chart 1: Correlation Heatmap
plt.figure(figsize=(8, 6))
heatmap_data = df_clean[['lifeExp', 'log_gdpPercap', 'pop', 'year']].corr()
sns.heatmap(heatmap_data, annot=True, cmap='RdBu_r', vmin=-1, vmax=1, linewidths=0.5)
plt.title('Global Correlation Matrix', fontsize=16)
plt.show()
```



Interpretation: The heatmap confirms a strong positive correlation (**0.81**) between `log_gdpPercap` and `lifeExp`. Interestingly, `year` also has a moderate positive correlation with life expectancy, reflecting global technological progress over time.

```
In [86]: # Chart 2: The "Hans Rosling" Animation
## If you encounter issues with Plotly rendering, ensure you have the conda
fig_2 = px.scatter(df_clean,
                  x="gdpPercap",
                  y="lifeExp",
                  animation_frame="year",
                  animation_group="country",
                  size="pop",
                  color="continent",
                  hover_name="country",
                  log_x=True,
```



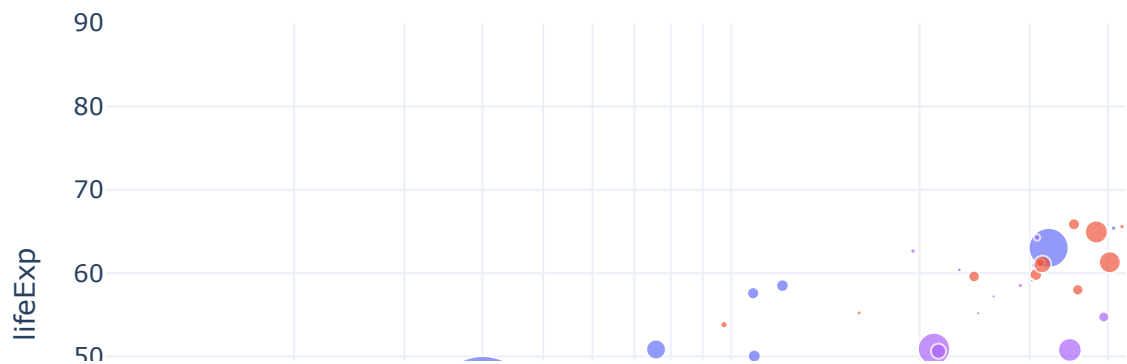
```

size_max=55,
range_x=[100, 100000],
range_y=[25, 90],
title="Evolution of Global Development (1952-2007)",
template="plotly_white")

fig_2.update_layout(title_font_size=20)
fig_2.show()
fig_2.write_html("Evolution of Global Development (1952-2007)_plot.html", ir

```

Evolution of Global Development (1952-2007)



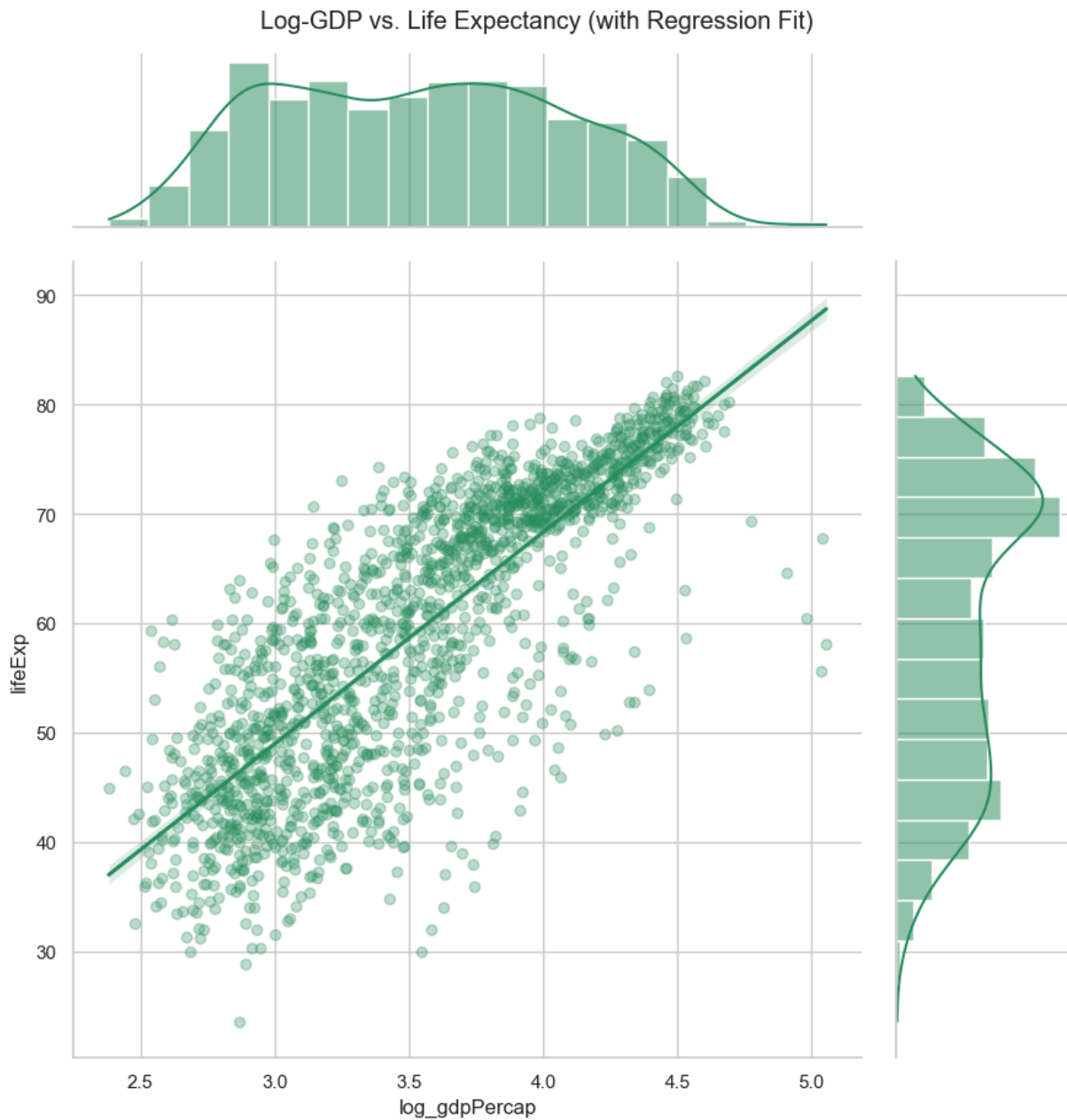
Interpretation:

- **1952:** Most African and Asian countries are clustered in the bottom-left (poor & short lives).
- **2007:** The entire world has moved "up and to the right." However, a distinct curve emerges: once a country passes ~\$5,000 GDP per capita, life expectancy gains slow down (diminishing returns).

Quantifying the Relationship (Static Regression)

The linear regression in Section 4 suggested a strong fit. Here, we visualize that model with a confidence interval.

```
In [87]: # Chart 3: Regression Plot with Marginal Distributions
g = sns.jointplot(data=df_clean, x="log_gdpPercap", y="lifeExp", kind="reg",
                  height=9, ratio=4, color="#2b905f", scatter_kws={'alpha':
g.fig.suptitle('Log-GDP vs. Life Expectancy (with Regression Fit)', y=1.02)
plt.show()
```



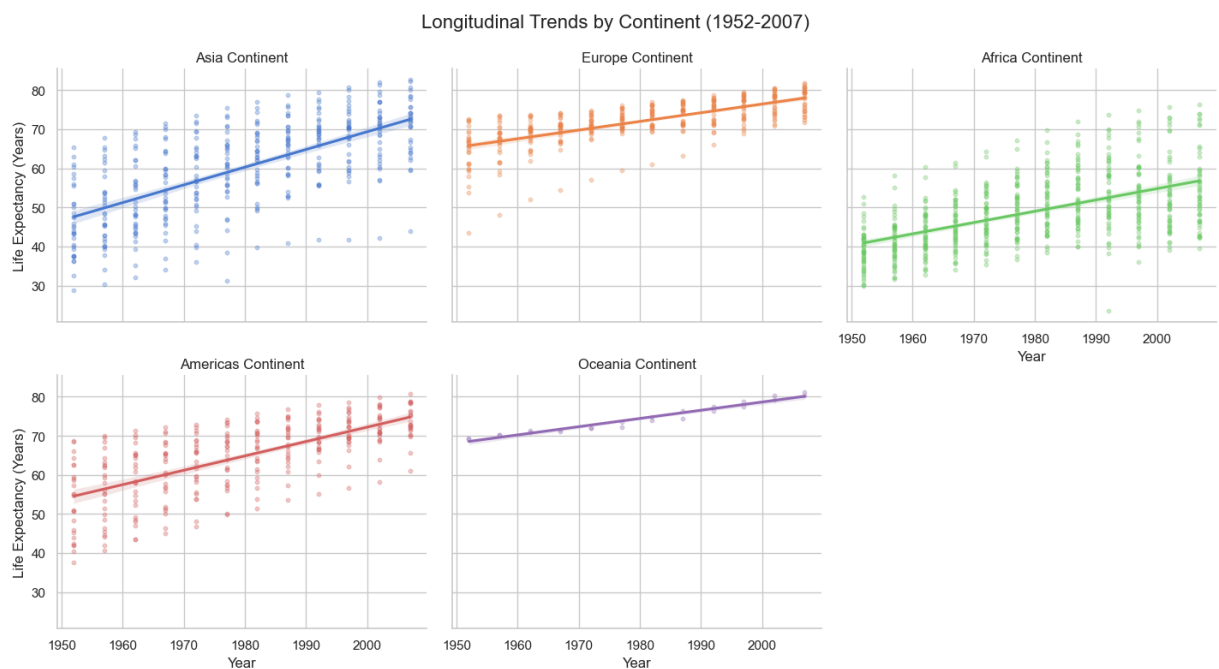
Interpretation: The regression line (solid) passes through the dense cluster of data points, confirming the model. The marginal histograms on the sides both show that Log-GDP and Life Expectancy are still not normal distribution.

Regional Divergence (Subplots)

Which continents are improving consistently? I use small multiples (Facet Grid) to separate the noise.

```
In [88]: # Chart 4: Regional Facet Grid
g = sns.lmplot(data=df_clean, x="year", y="lifeExp", col="continent", hue="c",
               col_wrap=3, height=4, aspect=1.2, scatter_kws={'alpha': 0.3},

# Add titles and adjust layout
g.set_titles("{col_name} Continent")
g.set_axis_labels("Year", "Life Expectancy (Years)")
plt.subplots_adjust(top=0.9)
g.fig.suptitle('Longitudinal Trends by Continent (1952-2007)', fontsize=16)
plt.show()
```



Interpretation:

- **Europe/Oceania:** Started high and stayed high with slow, steady growth.
- **Asia:** Shows the steepest slope, indicating the fastest "catch-up" (Rapid development).
- **Africa:** Shows a flattening trend (and even a dip in some countries) around the 1990s, visually capturing the stagnation discussed in the hypothesis.

Visualizing Growth Velocity (Interactive)

Using the results from the Pivot Table in Section 4, and visualize exactly how much each region grew.

```
In [89]: # Data from Section 4 Pivot Table
# Note: Ensure 'regional_progress' is defined from previous cells or recreated
regional_progress_viz = regional_progress.reset_index()

# Chart 5: Interactive Bar Chart of Growth
fig_4 = px.bar(regional_progress_viz,
                x='continent',
                y='growth_pct',
                color='growth_pct',
                text_auto='.1f',
                title='Total Percentage Growth in Life Expectancy (1952-2007)',
                labels={'growth_pct': 'Growth (%)', 'continent': 'Region'},
                color_continuous_scale='Viridis',
                template="plotly_white")

fig_4.update_layout(xaxis_title="Continent", yaxis_title="Percentage Growth")
fig_4.show()
fig_4.write_html("Total Percentage Growth in Life Expectancy (1952-2007)_plotly.html")
```

Total Percentage Growth in Life Expectancy (1952-2007)



Interpretation: Asia is the clear winner of the "Catch-Up" race with over **60% growth** in life expectancy, double that of the Americas. This confirms the "Asian Miracle" of the late 20th century.

The 2007 Inequality Snapshot

Finally, looking at the distribution of the most recent data to understand the current state of global inequality.

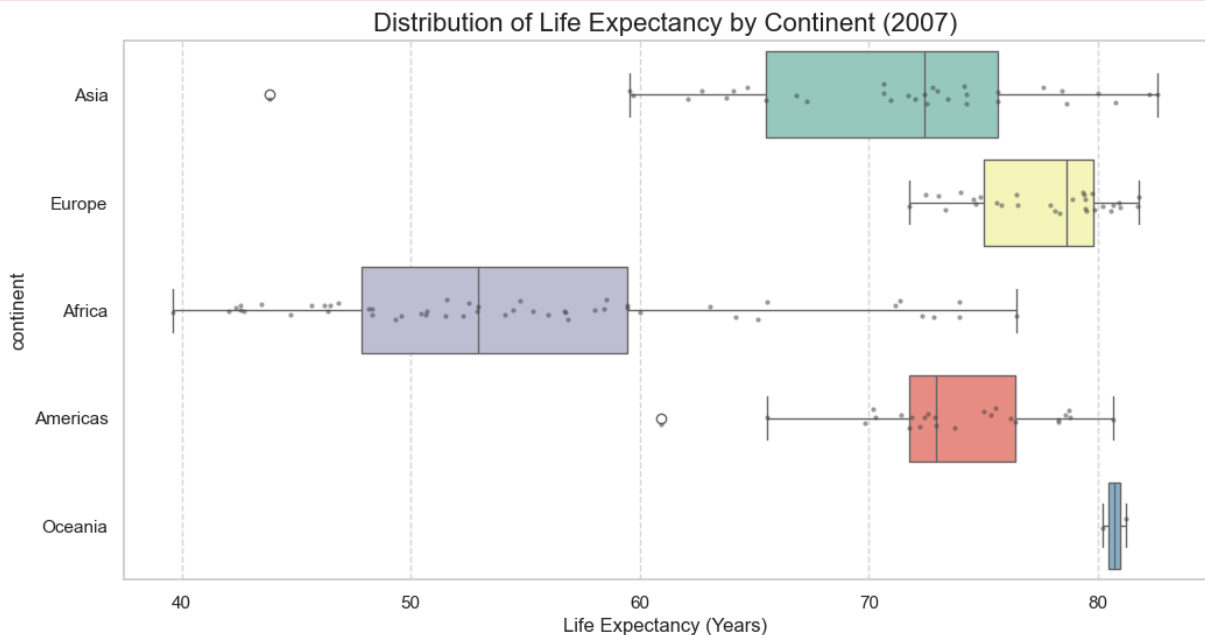
```
In [90]: # Chart 6: Boxplot of 2007 Data
plt.figure(figsize=(12, 6))
data_2007 = df_clean[df_clean['year'] == 2007]

sns.boxplot(x='lifeExp', y='continent', data=data_2007, palette="Set3")
sns.stripplot(x='lifeExp', y='continent', data=data_2007, color=".25", alpha=0.5)

plt.title('Distribution of Life Expectancy by Continent (2007)', fontsize=16)
plt.xlabel('Life Expectancy (Years)')
plt.grid(True, axis='x', linestyle='--', alpha=0.7)
plt.show()
```

```
/var/folders/fl/f_hjv3lj25z58v5bn3btbssh0000gn/T/ipykernel_11928/3229293207.py:5: FutureWarning:
```

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `y` variable to `hue` and set `legend=False` for the same effect.



Interpretation:

- **Africa** has the widest spread and the lowest median, confirming it is the region with the most internal inequality and challenges.
- **Europe and Oceania** are highly compacted at the top, meaning almost everyone in those regions enjoys a long life (low inequality).

Section 6 - Conclusions

6.1 Research Question Answers & Quantified Insights

This analysis successfully demonstrated the strong but non-linear relationship between economic output and human longevity.

- **RQ1 (The Wealth-Health Link):** A strong positive correlation ($r = 0.81$) was observed between `log_gdpPercap` and `lifeExp`. The regression model confirms a "diminishing returns" effect, i.e., while early economic gains drastically improve health, the benefits plateau after a nation reaches approximately **\$5,000 GDP per capita**.

- **RQ2 (Temporal Shifts):** The world is living longer. From 1952 to 2007, the global weighted average life expectancy rose significantly. The "Hans Rosling" visualization revealed that while the entire world moved "up and to the right," the **wealth gap** (skewness) remains persistent.
- **RQ3 (Regional Catch-Up):** **Asia** demonstrated the most remarkable velocity, achieving over **60% growth** in life expectancy, effectively "catching up" to the West. In contrast, **Africa** showed signs of stagnation in the 1990s (likely due to the HIV/AIDS epidemic), creating a distinct "negative tail" in the global distribution.

6.2 Limitations & Methodology Reflection

While the results are robust, the following limitations must be acknowledged:

- **Data Age:** The dataset concludes in **2007**. It fails to capture significant modern events like the 2008 financial crisis or the COVID-19 pandemic, limiting current predictive power.
- **Proxy Limitations:** Using GDP as a sole proxy for "wealth" ignores income inequality *within* countries (Gini coefficient). A country can be "rich" on average while its population remains poor.
- **Causality:** Correlation does not imply causation. While wealth helps health, healthier populations also produce more wealth (reverse causality).

6.3 Actionable Recommendations & Future Work

- **Policy Focus:** International health aid yields the highest marginal return in nations with GDP under **\$3,000**. Beyond this threshold, focus should shift to lifestyle-related health factors rather than basic economic aid.
- **Future Improvement:** A stronger model would incorporate **Education Index** and **Healthcare Spending** columns to create a Multiple Regression model, likely increasing our R^2 value beyond 0.65. (Explanability)

Section 7 - AI Log

Project Phase	Specific Usage Description
1. Ideation & Strategy	Consulted AI to refine the initial research questions, shifting focus from simple trends to "Regional Divergence" and "Wealth-Health Correlation" to meet the "Above and Beyond" criteria.
2. Data Cleaning	Used AI to generate the <code>detect_health_outliers</code> function and to verify the logic behind using a Logarithmic Transformation for the skewed GDP data.

Project Phase	Specific Usage Description
3. Coding & Debugging	Requested AI assistance to write the complex syntax for the Plotly Interactive Animation (Bubble Chart) and the Weighted Mean aggregation function, ensuring the code followed DRY (Don't Repeat Yourself) principles.
4. Analysis & Writing	Used AI to help structure the academic tone of the Introduction and Conclusion sections. The specific statistical interpretations (e.g., analyzing skewness shifts over time) were brainstormed with AI assistance.

I have reviewed all code suggestions provided by the AI to ensure I understand their functionality.

Export PDF

```
In [ ]: import plotly.io as pio
pio.renderers.default = "notebook_connected"

# python -m jupyter nbconvert --to webpdf --embed-images --allow-chromium-downloads
```

Export Interactive Plots

```
In [92]: import plotly.io as pio
figs = [fig_2, fig_4] # Add other figures as needed
parts = []
for i, fig in enumerate(figs):
    parts.append(
        pio.to_html(
            fig,
            full_html=False,
            include_plotlyjs="inline" if i == 0 else False, # 只在第一張內嵌
        )
    )
```